Simulation Arena

Deal Approval & Margin Balancing

Approve anything between 18% and 22% - 20% is the best single number. The 25% the rep is asking for is past it - that costs about $212.87M of expected profit.

Seed 42 | Run run_08ceab3ea71e | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 21:29

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
20%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$5,019.01M
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
18% to 22%
Across this range the expected profit is the same within the model's own margin of error. The model's own margin of error here is about $117.09M, so anywhere in this range is a judgement call, not a calculation.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
25%
At 25% the expected profit falls $212.87M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $212.87M

EXPECTED PROFIT IF YOU CAVE IN
$4,806.14M
What 25% actually earns you.
vs -$212.87M vs the optimum

PROFIT GIVEN AWAY
$212.87M
Buying +11 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
75.2%
The chance the deal lands.

WORST CASE (1-IN-20)
$0
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $209.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $10,066.51M on $20,911.98M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 20000 of your past deals

3. The sweet spot

20% discount gives the highest expected profit at $5,019.01M, with a 75.2% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

Across this range the expected profit is the same within the model's own margin of error. 20% is the best single number, but anything from 18% to 22% is a defensible call - that is where the rep has room to negotiate.

Evidence: margin of error $117.09M; safe range 18% to 22%

5. What the requested discount really costs

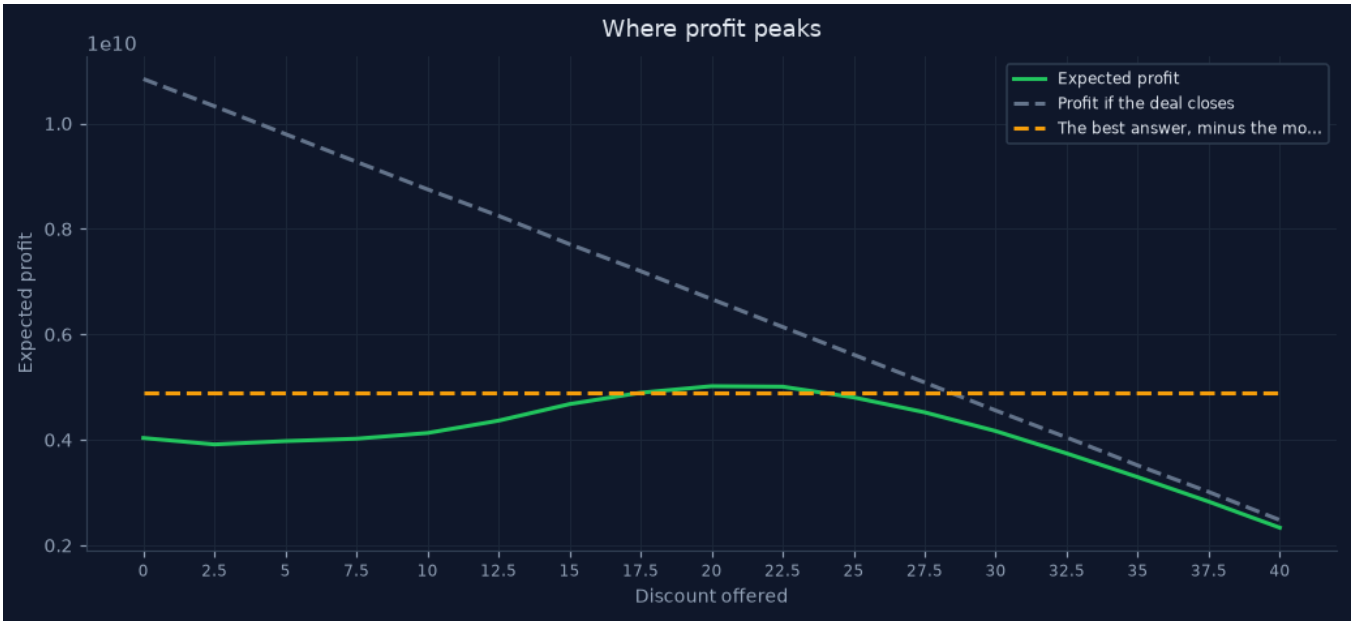
Moving from 22% to 25% buys +11 points of win probability but costs $212.87M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: $4,806.14M vs $5,019.01M

Charts

Where profit peaks

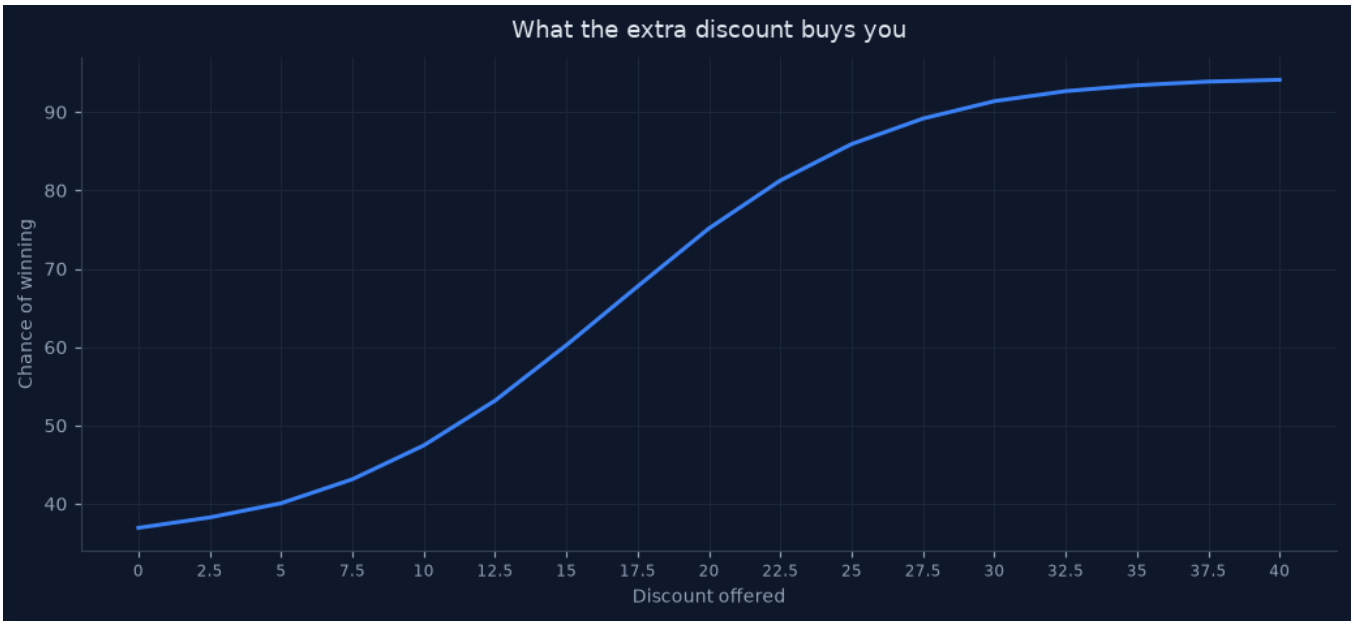
Everything from 18% to 22% gives the same answer within the model's margin of error; the peak is 20%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 18% to 22% range. Past that you are genuinely paying for the win.

What the extra discount buys you

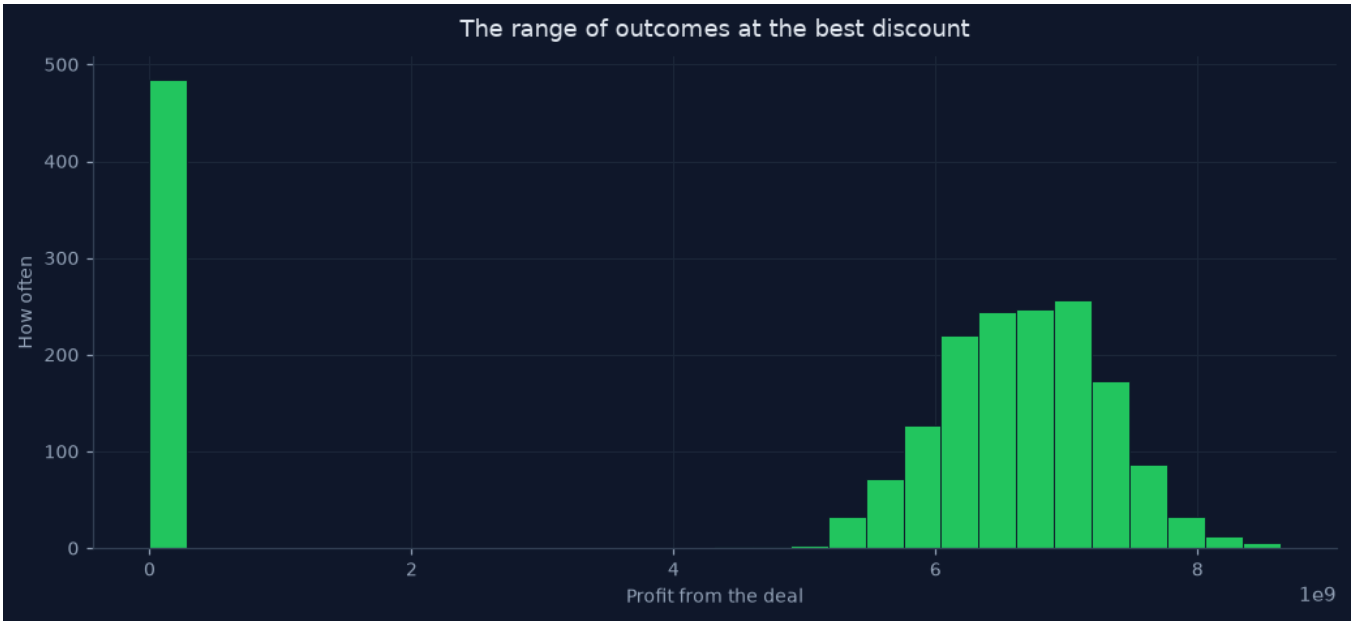
Curve learned from 20000 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 20% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$20,911,979,419	$10,844,362,252	36.9%	$4,036,200,261	$0	0% chance
2.5%	$20,389,179,933	$10,328,290,058	38.3%	$3,914,433,695	$0	0% chance
5.0%	$19,866,380,448	$9,799,392,137	40.1%	$3,977,789,758	$0	0% chance
7.5%	$19,343,580,962	$9,273,325,489	43.1%	$4,022,612,883	$0	0% chance
10.0%	$18,820,781,477	$8,751,718,732	47.5%	$4,131,040,611	$0	0% chance
12.5%	$18,297,981,991	$8,249,335,133	53.2%	$4,365,352,246	$0	0% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$17,775,182,506	$7,711,956,445	60.3%	$4,679,161,830	$0	0% chance
17.5%	$17,252,383,021	$7,200,823,919	67.8%	$4,894,896,927	$0	0% chance
20.0%	$16,729,583,535	$6,670,245,125	75.2%	$5,019,010,476	$0	0% chance
22.5%	$16,206,784,050	$6,142,640,099	81.3%	$5,009,235,404	$0	0% chance
25.0%	$15,683,984,564	$5,616,421,657	85.9%	$4,806,139,326	$0	0% chance
27.5%	$15,161,185,079	$5,089,635,609	89.2%	$4,523,056,644	$0	0% chance
30.0%	$14,638,385,593	$4,557,682,341	91.4%	$4,167,569,289	$0	0% chance
32.5%	$14,115,586,108	$4,045,946,605	92.7%	$3,743,590,143	$0	0% chance
35.0%	$13,592,786,622	$3,516,506,830	93.4%	$3,294,799,288	$0	0% chance
37.5%	$13,069,987,137	$3,012,988,073	93.9%	$2,829,376,517	$0	1% chance
40.0%	$12,547,187,651	$2,480,313,371	94.1%	$2,335,218,340	$0	5% chance

Row 9 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (20%)	Asked (25%)	Difference
Discount	20.0%	25.0%	+5.0 pts
Win chance	75.2%	85.9%	+10.8 pts
Profit if won	$6,670,245,125	$5,616,421,657	-$1,053,823,468
Expected profit	$5,019,010,476	$4,806,139,326	-$212,871,151
Range where the answer is the same	18% to 22%	outside it	margin of error $117.09M

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$4,036,200,261	$982,810,216	Outside	costs you $982.81M
2.5%	$3,914,433,695	$1,104,576,781	Outside	costs you $1,104.58M
5.0%	$3,977,789,758	$1,041,220,719	Outside	costs you $1,041.22M
7.5%	$4,022,612,883	$996,397,593	Outside	costs you $996.40M
10.0%	$4,131,040,611	$887,969,866	Outside	costs you $887.97M
12.5%	$4,365,352,246	$653,658,231	Outside	costs you $653.66M
15.0%	$4,679,161,830	$339,848,646	Outside	costs you $339.85M
17.5%	$4,894,896,927	$124,113,549	Inside	safe to approve
20.0%	$5,019,010,476	$0	Inside	safe to approve
22.5%	$5,009,235,404	$9,775,073	Inside	safe to approve
25.0%	$4,806,139,326	$212,871,151	Outside	costs you $212.87M
27.5%	$4,523,056,644	$495,953,833	Outside	costs you $495.95M
30.0%	$4,167,569,289	$851,441,188	Outside	costs you $851.44M
32.5%	$3,743,590,143	$1,275,420,333	Outside	costs you $1,275.42M
35.0%	$3,294,799,288	$1,724,211,188	Outside	costs you $1,724.21M
37.5%	$2,829,376,517	$2,189,633,959	Outside	costs you $2,189.63M
40.0%	$2,335,218,340	$2,683,792,136	Outside	costs you $2,683.79M

Highlighted rows are the range where the answer is the same (18% to 22%) - within about $117.09M of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
Q-000011589	Acct-226235	$24,000,000	-5.7%	0%	-$502,859
Q-000009396	Acct-223884	$24,000,000	3.8%	0%	$337,013
Q-000009047	Acct-090640	$24,000,000	4.8%	0%	$424,814
Q-000008091	Acct-110274	$19,296,863	15.4%	0%	$1,096,718
Q-000007533	Acct-062342	$16,052,463	0.7%	0%	$40,337
Q-000018826	Acct-138872	$14,565,883	23.4%	0%	$1,260,595
Q-000002411	Acct-019732	$14,340,308	-11.8%	0%	-$623,716
Q-000006273	Acct-125458	$14,145,541	46.0%	20%	$2,766,932
Q-000005246	Acct-233590	$14,094,974	12.6%	0%	$654,193
Q-000013379	Acct-205409	$13,878,192	21.0%	0%	$1,078,510
Q-000004222	Acct-219531	$13,822,773	2.4%	0%	$123,102
Q-000009736	Acct-113136	$13,452,114	-3.5%	0%	-$172,493
Q-000017300	Acct-007220	$13,422,614	25.1%	0%	$1,245,477
Q-000000139	Acct-108061	$13,011,806	-15.3%	0%	-$736,628
Q-000009880	Acct-201403	$12,782,137	-22.3%	0%	-$1,053,320
Q-000001755	Acct-058430	$12,706,362	-8.4%	0%	-$393,945
Q-000001411	Acct-122168	$12,567,749	23.2%	0%	$1,077,915
Q-000010633	Acct-176419	$12,154,931	1.3%	0%	$59,290
Q-000013798	Acct-106373	$11,654,026	18.1%	0%	$779,052
Q-000010069	Acct-070297	$11,489,319	15.7%	0%	$665,721
Q-000013273	Acct-130561	$11,410,414	27.0%	0%	$1,136,772
Q-000007138	Acct-039861	$11,334,211	18.2%	0%	$763,956
Q-000005723	Acct-178920	$11,250,338	16.4%	0%	$680,975
Q-000001338	Acct-100033	$11,225,300	17.8%	0%	$739,607
Q-000009264	Acct-188657	$11,173,578	22.0%	0%	$908,792

What we recommend

Approve anything between 18% and 22% - 20% is the best single number. The 25% the rep is asking for is past it - that costs about $212.87M of expected profit.

Risk Minimal 0 out of 100 | Confidence: High

Why

• Any discount from 18% to 22% gives the same expected profit within the model's own margin of error, so there is real room to negotiate there.

• Expected profit peaks at 20% ($5,019.01M).

• The requested 25% delivers $4,806.14M - $212.87M less than the peak, which is more than the model's margin of error.

• Win probability only improves +11 points for that giveaway.

What to do

1. Approve any discount between 18% and 22% - they are the same answer within the model's own margin of error.

2. Above 22% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $85.15M per point of discount.

3. If this quarter must not misfire, 0% is the safer call: it maximises the bad case rather than the average.

The trade-off: You are trading $212.87M of expected profit for +11 points of win probability.

What we assumed and what to watch

Assumptions

• List value $20,911,979,419, cost $10,066,512,144 (55.0% standard margin).

• Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.6% discount (source: 20000 of your past deals).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $117.09M at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 18% to 22%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> your 'deal_id' column

• Customer name -> your 'customer' column

• Deal value (list price) -> your 'deal_value' column

• Cost to deliver -> your 'cost' column

• Standard margin (%) -> your 'margin_pct' column

• Discount given (%) -> your 'discount_pct' column

• Won? (1/0) -> your 'won' column

• Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column

• Unmetered burst used -> your 'burst_allowance_pct' column

• Rollover window -> your 'rollover_window_months' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Unmetered burst used -> your 'burst_allowance_pct' column
0.00s	prepare	Rollover window -> your 'rollover_window_months' column
0.76s	prepare	Read 20,000 records into the model.
0.76s	simulate	Running the simulation…
0.76s	read	Reading the deal you gave us...
0.76s	read	You typed the deal size or the margin yourself, so we are using the figures you entered rather than reading them from your file.
0.76s	validate	You typed the margin, so we used your figure rather than the file's ($20,911.98M at 9.4%).
0.76s	read	Found 20000 deal(s) worth $20,911.98M at an average 55.0% margin.
0.76s	prepare	Cost to deliver is about $9,410.39M.
0.77s	prepare	Added the consumption effects from your file: $656.12M of extra cost to serve.
0.77s	prepare	Working out how much a discount really moves the buyer...

Time	Stage	What happened
1.12s	prepare	Learned the buyer's behaviour from 20000 past deals (they flip hardest around 16.6% discount).
1.12s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...
1.12s	simulate	Simulated 11% of the discount levels...
1.12s	simulate	Simulated 23% of the discount levels...
1.12s	simulate	Simulated 35% of the discount levels...
1.12s	simulate	Simulated 47% of the discount levels...
1.13s	simulate	Simulated 58% of the discount levels...
1.13s	simulate	Simulated 70% of the discount levels...
1.13s	simulate	Simulated 82% of the discount levels...
1.13s	simulate	Simulated 94% of the discount levels...
1.13s	simulate	Simulated 100% of the discount levels...
1.13s	summarize	All discount levels tested. Looking for the sweet spot...
1.13s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 18% to 22%.
1.15s	summarize	Done. Best discount is 20% ($5,019.01M expected profit); the range where the answer is the same is 18% to 22%.